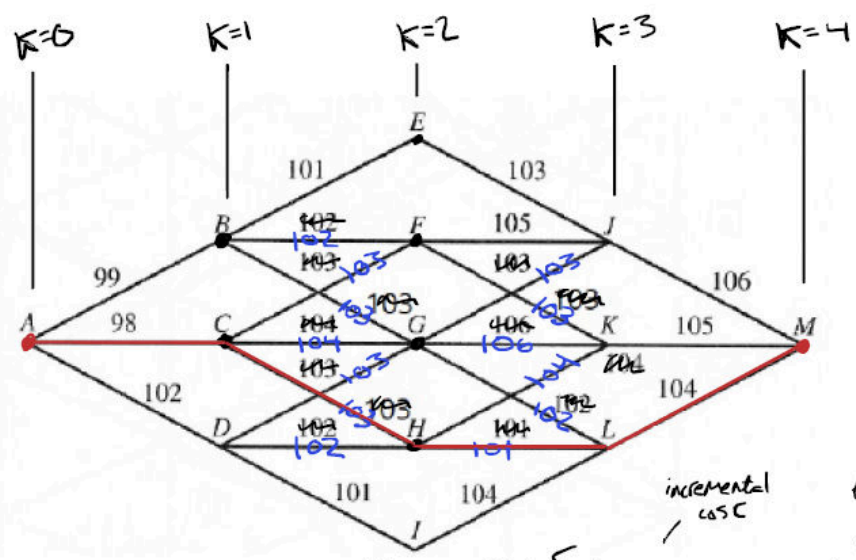
ECE675 HW3 Problem 2:

ECE675
HW3
Problem 2
Dynam. Prog.



incremental cost

terminal cost

$$V_k = \min_{u \in U} \left\{ L(k,x,u) + V_{k+1}\left(F(k,x,u)\right) \right\}$$

@ K=4

$V_4(M) = 0 + 0$ (no terminal cost)

@ K=3

$V_3(J) = 106 \quad V_3(K) = 105 \quad V_3(L) = 104$

@ K=2

$V_2(E) = 103 + V_3(J) = 209$

$V_2(F) = \min\{105 + V_3(J), 103 + V_3(K)\} = 208$

$V_2(G) = \min\{103 + V_3(J), 106 + V_3(K), 102 + V_3(L)\} = 206$

$V_2(H) = \min\{104 + V_3(K), 101 + V_3(L)\} = 205$

$V_2(I) = 104 + V_3(L) = 208$

@ K=1

$V_1(B) = \min\{101 + V_2(E), 102 + V_2(F), 103 + V_2(G)\} = 309$

$V_1(C) = \min\{103 + V_2(F), 104 + V_2(G), 103 + V_2(H)\} = 308$

$V_1(D) = \min\{103 + V_2(G), 102 + V_2(H), 101 + V_2(I)\} = 307$

@ K=0

$V_0(A) = \min\{99 + V_1(B), 98 + V_1(C), 102 + V_1(D)\} = 406$

- Minimum cost = 406
- Best route highlighted red

ECE675 HW3 Problem 3:

Problem 3 The inverted pendulum system consists of an inverted pendulum mounted on a cart shown in Fig.2.

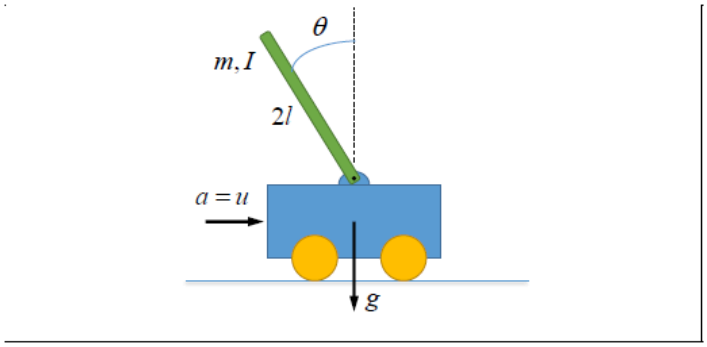


Figure 2: Inverted pendulum on a cart

The dynamics of the inverted pendulum are modeled in state-space form as follows:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -a_0 \sin x_1 - a_1 x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ b_0 \cos x_1 u \end{bmatrix}$$

where $x = [x_1\, x_2]^T = [\theta\, \dot{\theta}]^T$, and $a_0 = -\frac{mgl}{I+ml^2}$, $a_1 = \frac{b}{I+ml^2}$, and $b_0 = \frac{ml}{I+ml^2}$. For the considered system, the parameters are: $a_0 = -2.5$, $a_1 = 0.05$, and $b_0 = 0.25$.

(1) Determine the linearized dynamics about the equilibrium $x = [0\, 0]^T$.

(2) Convert the continuous-time dynamics to discrete-time using $\Delta t = 0.1$.

(3) Design a feedback control law $u = -Kx$ using Matlab command `dlqr` for the discretized system in (2). Set $Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $R = 1$. Simulate the inverted pendulum system in closed-loop with the designed controller and plot the results of the simulation.

```
x0 = [pi/4; 0];
sim('Prob3a.mdl',[0 10])

figure
subplot(2,1,1)
plot(x.time,x.signals.values)
ylim([-pi/2 pi/2])
legend('\theta','d\theta/dt')
grid on
subplot(2,1,2)
plot(u.time,u.signals.values)
ylim([-pi pi])
grid on
```

Problem 3.1: Linearized Dynamics

```
syms a0 a1 b0 x1 x2

% Linearize at x = [0; 0], cos(x1) --> 1, sin(x1) --> x1
A_lin = [0, 1; -a0, -a1]
```

A_lin =

$$\begin{pmatrix} 0 & 1 \\ -a_0 & -a_1 \end{pmatrix}$$

```
B_lin = [0; b0]
```

B_lin =

$$\begin{pmatrix} 0 \\ b_0 \end{pmatrix}$$

```
% Substitute in the numerical given values:
A_lin = double(subs(A_lin,[a0,a1,b0],[-2.5,0.05,0.25]))
```

```
A_lin = 2×2
         0    1.0000
    2.5000   -0.0500
```

```
B_lin = double(subs(B_lin,[a0,a1,b0],[-2.5,0.05,0.25]))
```

```
B_lin = 2×1
         0
    0.2500
```

Problem 3.2: Discretize the Dynamics:

```
% Sample Rate
ts = 0.1;
% Intitial Cond.
x0 = [pi/4; 0];

% Define continuous system, build discrete ss system
ContLinSystem = ss(A_lin,B_lin,zeros(2,2),zeros(2,1))
```

```
ContLinSystem =

  A =
           x1     x2
   x1       0      1
   x2     2.5  -0.05

  B =
         u1
   x1     0
   x2  0.25

  C =
       x1  x2
   y1   0   0
   y2   0   0

  D =
       u1
   y1   0
   y2   0

Continuous-time state-space model.
Model Properties
```

```
DiscLinSystem = c2d(ContLinSystem,ts);

% Extract A & B for discrete system
Ad_lin = DiscLinSystem.A
```

```
Ad_lin = 2×2
```

```
    1.0125    0.1002
    0.2504    1.0075
```

```
Bd_lin = DiscLinSystem.B
```

```
Bd_lin = 2×1
    0.0013
    0.0250
```

```
Cd_lin = eye(2,2)
```

```
Cd_lin = 2×2
     1     0
     0     1
```

```
Dd_lin = DiscLinSystem.D
```

```
Dd_lin = 2×1
     0
     0
```

Problem 3.3: Discrete LQR Controller

```
Q = [1 0; 0 0];
R = 1;

[K_dlqr, S, P] = dlqr(Ad_lin,Bd_lin,Q,R)
```

```
K_dlqr = 1×2
   18.5997   11.5669
S = 2×2
10^3 ×
    1.2961    0.8036
    0.8036    0.4997
P = 2×1 complex
   0.8535 + 0.0064i
   0.8535 - 0.0064i
```

```
simout1 = sim('ECE675_HW3_Problem3_sim.mdl',[0 10])
```

```
simout1 
  Simulink.SimulationOutput:

                logsout: [1x1 Simulink.SimulationData.Dataset]
                   tout: [101x1 double]

     SimulationMetadata: [1x1 Simulink.SimulationMetadata]
           ErrorMessage: [0x0 char]
```

```
theta = simout1.logsout{1}.Values.Data(:,1);
theta_dot = simout1.logsout{1}.Values.Data(:,2);
time = simout1.logsout{1}.Values.Time(:,1);
input = simout1.logsout{2}.Values.Data(:,1);

figure
subplot(2,1,1)
```

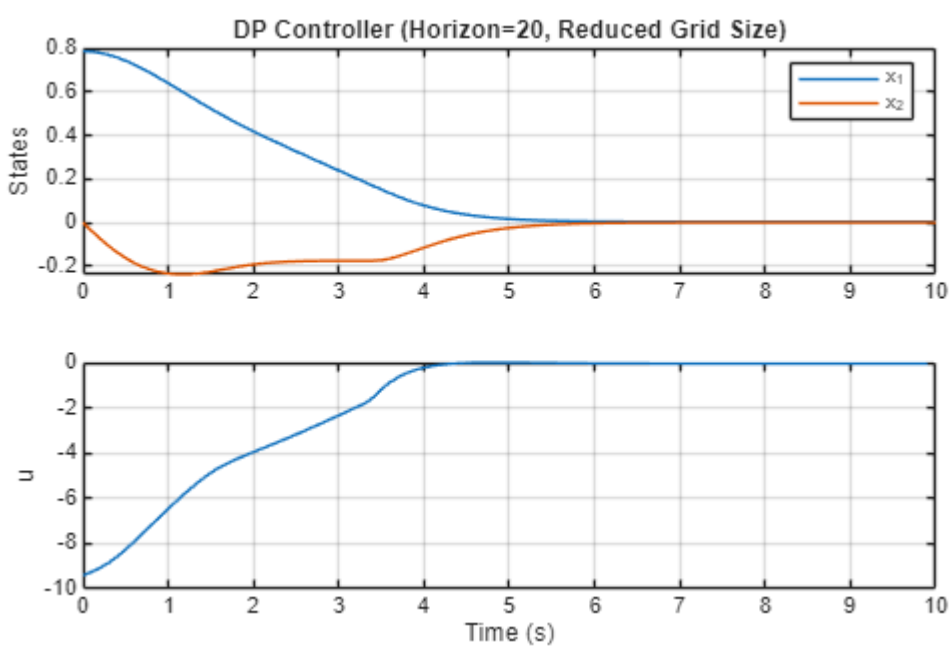
```
plot(time,theta)
hold on
plot(time, theta_dot)
legend('\theta','d\theta/dt')
grid on
subplot(2,1,2)
plot(time, input)
ylim([-10,pi])
legend('Input u')
title('dlqr Controller')
grid on
```

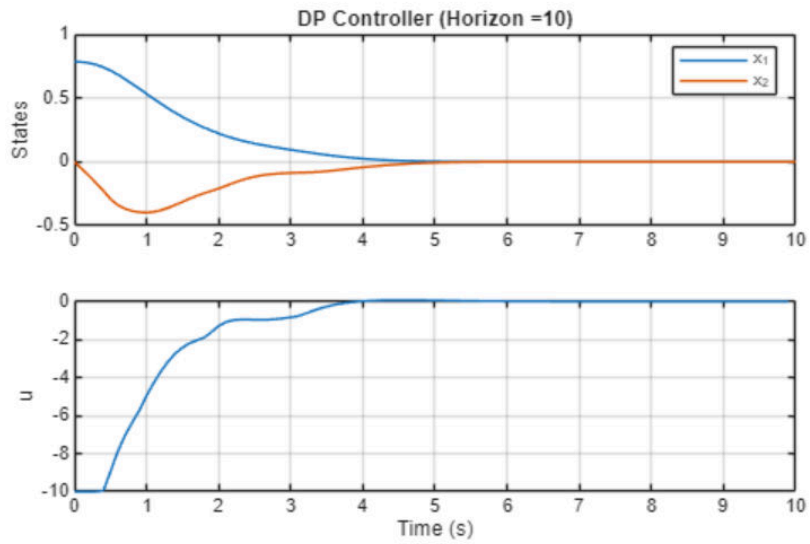
Problem 3.4: Dynamic Program Control Law

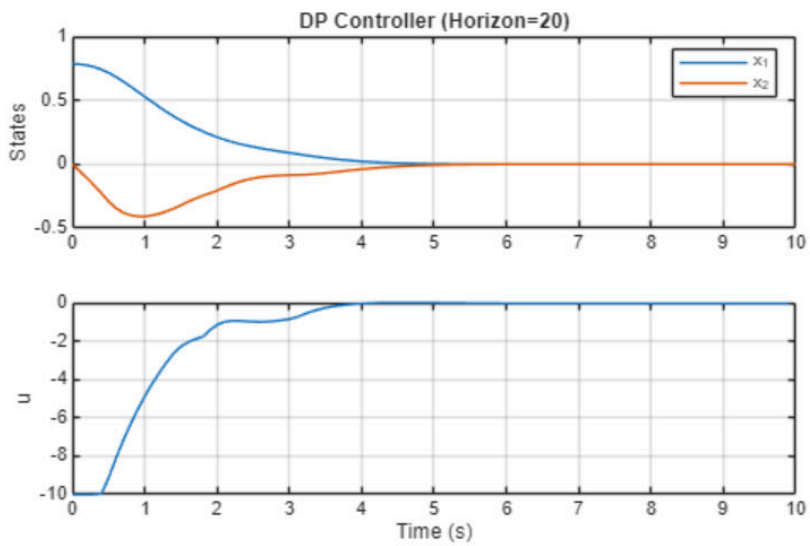
```
% Get discrete LQR terminal cost matrix using dlqr (you already computed)
% [K_dlqr, S, ~] = dlqr(Ad_lin,Bd_lin,Q,R);   % S is solution of discrete riccati
(use your existing S)

% params
X1GridSize = 10;
X2GridSize = X1GridSize;
DP_horizon = 20;
Q = [1 0; 0 0];
R = 1;
ts = 0.1;

% stage + terminal cost
Jstage = @(x,u) x'*Q*x + u'*R*u;
Jtogo = @(x) x'*S*x;

% state grid
X1 = linspace(-pi/2, pi/2, X1GridSize);
X2 = linspace(-pi/2, pi/2, X2GridSize);

% initialize tables for cost & optimal inputs
U = zeros(X1GridSize, X2GridSize);
J  = zeros(X1GridSize, X2GridSize);

% bounded u search grid
umax = 10; % control bounds should be within this pretty easily
u_grid = linspace(-umax, umax, 201);

% Value iteration
for iter = 1:DP_horizon
    for i = 1:length(X1)
        for j = 1:length(X2)
            x = [X1(i); X2(j)];
```

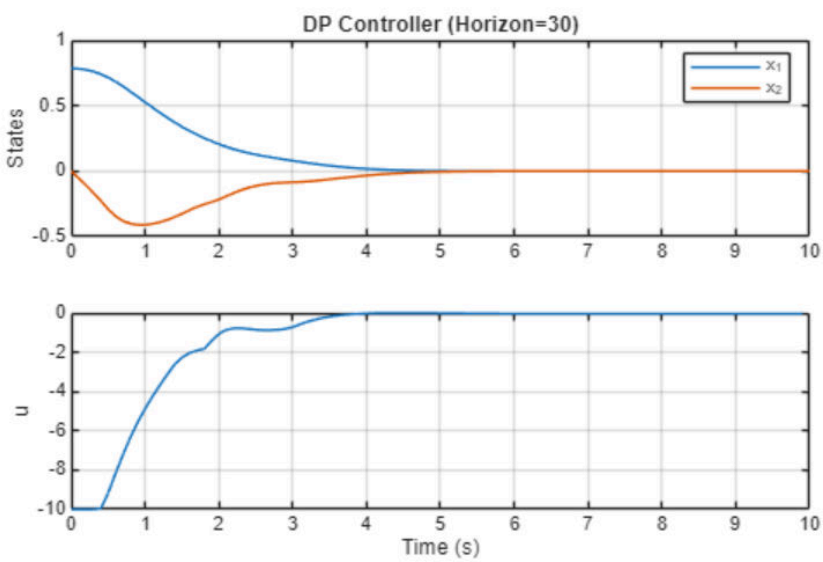
```
            % evaluate cost for every u candidate
            % next state
            Xnext = Ad_lin*x + Bd_lin * u_grid;   % 2 x numU
            % compute cost: stage + terminal-on-next
            cost_vals = zeros(size(u_grid));
            for m = 1:numel(u_grid)
                xn = Xnext(:,m);
                % evaluate Jtogo at xn via interp2 on J table (if iter>1)
                if iter==1
                    future_cost = Jtogo(xn);  % currently x'*S*x
                else
                    % safe interp: clamp xn to grid before interp
                    x1c = min(max(xn(1), min(X1)), max(X1));
                    x2c = min(max(xn(2), min(X2)), max(X2));
                    future_cost = interp2(X2, X1, J, x2c, x1c, 'linear');
                end
                cost_vals(m) = Jstage(x, u_grid(m)) + future_cost;
            end

            % pick best u
            [mincost, midx] = min(cost_vals);
            U(i,j) = u_grid(midx);
            J(i,j) = mincost;
        end
    end

    % update Jtogo to the just-computed J table
    % NOTE: J is arranged as J(i over X1, j over X2)
    Jtogo = @(xq) interp2(X2, X1, J, ...
                          min(max(xq(2), min(X2)), max(X2)), ...
                          min(max(xq(1), min(X1)), max(X1)), 'linear');
end

% DP policy accessor (with clamping)
function u = GetInput_DP(x_state, X1, X2, U)
    x1 = x_state(1);
    x2 = x_state(2);
    % clamp
    x1c = min(max(x1, min(X1)), max(X1));
    x2c = min(max(x2, min(X2)), max(X2));
    u = interp2(X2, X1, U, x2c, x1c, 'linear');
end

GetOptimalInput = @(x_state) GetInput_DP(x_state, X1, X2, U);

%% === Simulate closed-loop with DP policy ===
Tfinal = 10;
timesteps = round(Tfinal/ts);
```

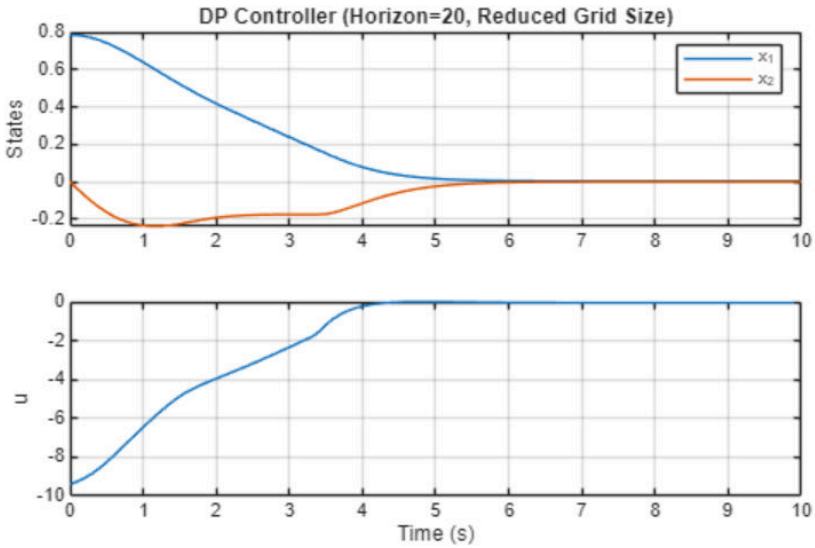
```
x_cl_DP = zeros(2, timesteps+1);
u_cl_DP = zeros(1, timesteps);

x_cl_DP(:,1) = x0;

for k = 1:timesteps
    curx = x_cl_DP(:,k);
    u = GetOptimalInput(curx);
    u_cl_DP(k) = u;
    x_cl_DP(:,k+1) = Ad_lin*curx + Bd_lin*u;
end

t = 0:ts:Tfinal;

% plots
figure;

subplot(2,1,1)
plot(t, x_cl_DP(1,:), t, x_cl_DP(2,:))
grid on
legend('x_1','x_2')
title('DP Controller (Horizon=20, Reduced Grid Size)')
ylabel('States')

subplot(2,1,2)
plot(t(1:end-1), u_cl_DP)
grid on
ylabel('u')
xlabel('Time (s)')
```




DP Controller (Horizon =10)
States
u
Time (s)
x1
x2


DP Controller (Horizon=20)
States
u
Time (s)
x1
x2


DP Controller (Horizon=30)
States
u
Time (s)
x1
x2



(5) Design an MPC-based feedback control law using a Matlab `quadprog` and compare the results with those in (4).

Closed loop response with quadprog is very similar to response using the DP control law with reduced grid size (10x10 grid). Increased grid size on the DP controller significantly reduces the control effort to bring the pendulum to upright position.

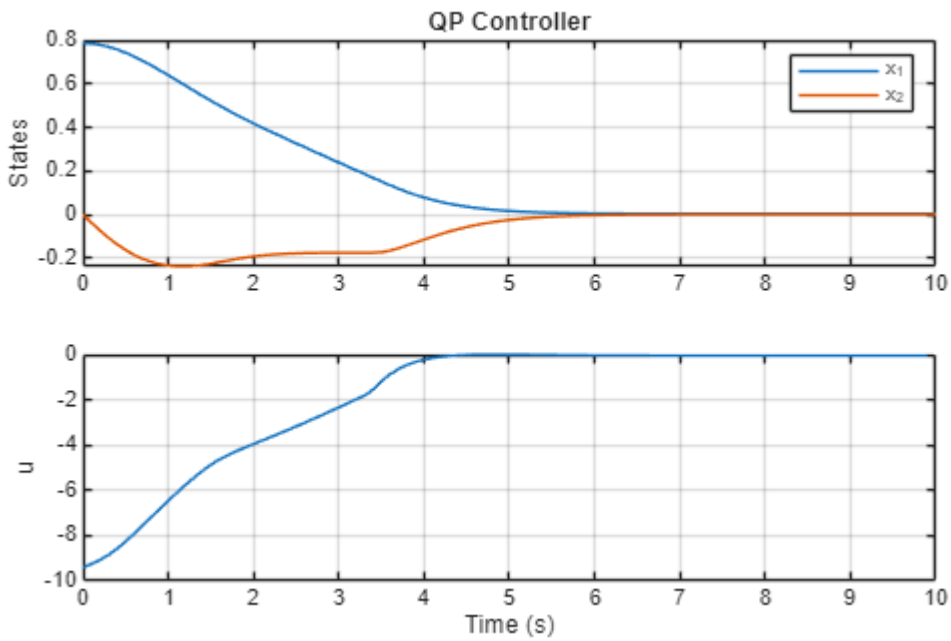
```
function u_opt = GetOptimalInputQP(x, Ad, Bd, Q, R, P)

% state-dependent
Ax = Ad * x;

% quad term
H = Bd' * P * Bd + R;
H = (H + H')/2;

% linear term
f = (Bd' * P * Ax);

% solve QP
u_opt = quadprog(H, f, [], [], [], [], [], [], [], ...
                 optimoptions('quadprog','Display','off'));

end

DP_horizon = 20;
Q = [1 0; 0 0];
R = 1;
P = zeros(2,2)
```

```
P = 2×2
     0     0
     0     0
```

```
ts = 0.1;

GetOptimalInputQPAnon = @(x) GetOptimalInputQP(x, Ad, Bd, Q, R, P)
```

```
GetOptimalInputQPAnon = function_handle with value:
    @(x)GetOptimalInputQP(x,Ad,Bd,Q,R,P)
```

```
Tfinal = 10;
timesteps = round(Tfinal/ts);

x_cl_QP = zeros(2, timesteps+1);
u_cl_QP = zeros(1, timesteps);

x_cl_QP(:,1) = x0;

for k = 1:timesteps
    curx = x_cl_QP(:,k);
    u = GetOptimalInput(curx);
    u_cl_QP(k) = u;
    x_cl_QP(:,k+1) = Ad_lin*curx + Bd_lin*u;
end

t = 0:ts:Tfinal;

% plots
figure;

subplot(2,1,1)
plot(t, x_cl_QP(1,:), t, x_cl_QP(2,:))
grid on
legend('x_1','x_2')
title('QP Controller')
ylabel('States')

subplot(2,1,2)
plot(t(1:end-1), u_cl_QP)
grid on
ylabel('u')
xlabel('Time (s)')
```


QP Controller
States
0.8
0.6
0.4
0.2
0
-0.2
0
1
2
3
4
5
6
7
8
9
10
x1
x2
u
0
-2
-4
-6
-8
-10
Time (s)